# Finding Order in Metric Structures

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# Metric Structures

## Definition

A *metric language* is just like a regular first-order language, consisting of functions and relations.

## Definition

A *metric structure* consists of:

- A complete metric space of diameter $\leq 1$
- For each $n$-ary function symbol, a uniformly continuous function $M^n \to M$
- For each $n$-ary relation symbol, a uniformly continuous function $M^n \to [0, 1]$

# Formulas

## Definition

An *atomic formula* is defined as usual, except instead of $=$, the basic relation is $d(x, y)$.

## Definition

A *formula* is

- An atomic formula
- $u(\phi_1, \ldots, \phi_n)$ where $\phi_i$s are formulas and $u : [0,1]^n \to [0,1]$ is continuous
- $\sup_x \phi$ or $\inf_x \phi$

# Making Linear Orders Metric

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# Making Linear Orders Metric

- Most famous examples of metric structures are stable or not NIP - how do we put an order on one?
- Itaï Ben Yaacov has described Ordered Real Closed Metric Valued Fields, but making the metric space bounded complicated things.
- Diego Bejarano and I are working to simplify this approach.

# Metric Linear Orders

- Call $M$ a *metric linear order* if
  - $M$ has a complete metric of diameter $\leq 1$
  - $M$ has a linear order
  - open balls are order-convex.
- $M$ is a metric structure in the language $\{r\}$, with

$$r(x,y) = \begin{cases} 0 & x \leq y \\ d(x,y) & y \leq x \end{cases}$$

- Think of $r(x,y)$ as "the amount $x$ is greater than $y$," or

$$r(x,y) = d\left(x, (\infty, y]\right).$$

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- $\sup_{x,y,z} r(x,z) \dot{-} (r(x,y) + r(y,z)) = 0$

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- DLO and $\mathrm{Th}(\mathbb{Z}, <)$ are two useful completions of the theory of linear orders
- We seek analogous completions of MLO
- For simplicity, assume the metric is an ultrametric.

# Axiomatizing Ultrametric Dense Linear Orders

**Definition**

Let UDLO be the theory of *ultrametric-dense linear orders*, consisting of MLO with the following axioms:

- $d(x,z) \leq \max(d(x,y), d(y,z))$
- For any rational $p \in \mathbb{Q} \cap [0,1]$, $\sup_x \inf_y |r(x,y) - p| = 0$
- For any rational $p \in \mathbb{Q} \cap [0,1]$, $\sup_x \inf_y |r(y,x) - p| = 0$.

Basically, the distances from $x$ to $y > x$ are dense in $[0,1]$.

# Stable Diversion: Dense Ultrametrics

Call an ultrametric space *dense* if the set of distances to any point is dense in $[0, 1]$.

**Fact (Conant)**

*The theory of dense ultrametrics is complete and has QE, but is not $\aleph_0$-categorical.*

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**Lemma (A., Bejarano)**

*The theory of dense ultrametrics is the model companion of the theory of ultrametrics, and is approximately $\aleph_0$-categorical.*

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**Theorem (A., Bejarano)**

*In a model of* UDLO,

- *the metric and order topologies agree*
- $\mathrm{dcl}$ *is closure*

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**Lemma (A., Bejarano)**

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**Theorem (Van Thé)**

*$U_S$ has extremely amenable automorphism group, following from Fraïssé theory in the* **discrete-logic** *language of ordered $S$-valued metric spaces.*

# *o*-Minimality in Discrete Logic

**Fact**

*If $M$ expands a linear order, TFAE:*

- *every formula $\phi(x)$ in one variable is qf-definable in $\{<\}$*
- *every formula $\phi(x)$ in one variable is a finite union of intervals.*

- If these happen, $M$ is *o-minimal*.

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- If these happen, $M$ is *o-minimal*.
- How do we describe these properties for MLOs?

# Metric *o*-Minimality

**Theorem (A., Bejarano)**

*If $M$ expands a metric linear order, TFAE:*

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**Definition**

If $M$ expands a metric linear order, call $M$ *o-minimal* if every formula $\phi(x)$ in one variable satisfies these equivalent properties.

# Metric $o$-Minimality

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**Definition**

If $M$ expands a metric linear order, call $M$ *$o$-minimal* if every formula $\phi(x)$ in one variable satisfies these equivalent properties.

By QE, a model of UDLO is $o$-minimal.

# Regulated Functions

Regulated functions $[a, b] \to \mathbb{R}$ were defined by Bourbaki.

**Lemma (A. Bejarano)**

*If $M$ is a linear order, $f : M \to [0, 1]$, TFAE:*

1. *$f$ is a uniform limit of step functions*
2. *For any $a < b$, $M$ can be partitioned into finitely many intervals on which either $f(x) > a$ or $f(x) < b$.*

*If $M \models$ UDLO, and $f : M \to M$ is definable ($d(f(x), y)$ is a formula), then $f$ is continuous and satisfies (2).*

# Metric Valued Fields

## Definition

A *metric valued field* is a field $K$ with an absolute value $|\cdot| : K \to \mathbb{R}^{\geq 0}$ satisfying valuation axioms:

- $|x| = 0 \iff x = 0$
- $|xy| = |x||y|$
- $|x + y| \leq \max(|x|, |y|)$

which gives rise to an ultrametric $d(x, y) = |x - y|$.

# Real Closed Metric Valued Fields

**Definition**

If $K$ is a metric valued field equipped with an order, $K \equiv \mathbb{R}$ as ordered fields in classical logic, and $|\cdot|$ takes values outside $\{0, 1\}$, call $K$ a *real closed metric valued field*.

These are almost metric structures, but the metric is unbounded, a problem for continuous logic.

# Metric Valued Fields, à Rideau-Kikuchi, Scanlon, Simon

- Typically we just deal with the ball $B_{\leq 1}(0)$.
- For a metric valued field $K$, $\{x \in K : |x| \leq 1\}$ is a subring - the *valuation ring*.
- This is a metric structure in the ring language.

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- This is a metric structure in the ring language.
- Unfortunately, these are elementarily equivalent to other convex subrings of metric valued fields.

# Ordered Real Closed Metric Valuation Rings

**Theorem (A., Bejarano)**

- *Models of our theory* ORCMVR *are convex subrings of real closed metric valued fields.*
- *These have quantifier-elimination once we add a divisibility relation.*
- *These can only be weakly o-minimal.*

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**Theorem (A., Bejarano)**

*Any model of* ORCMVF *(the projective line) is an o-minimal expansion of a model of* UDCO*, the complete theory of ultrametric dense cyclic orders.*

Thank you, Rutgers!